

SEC2_APM1110-FA2-DABU, L.

GitHub: <https://github.com/dabulira20/Prob-ProbDi-FAs/tree/main/FA3>

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1. Problem (#5)

A geospatial analysis system has four sensors supplying images. The percentage of images supplied by each sensor and the percentage of relevant images (relative to a query) from each sensor are given below.

Sensor	% of Images Supplied	% of Relevant Images
1	15	50
2	20	60
3	25	80
4	40	85

We are asked for the **overall percentage of relevant images**.

Solution Let B_i be the event that an image comes from sensor i , and let A be the event that an image is relevant. We are given

$$P(B_1) = 0.15, P(B_2) = 0.20, P(B_3) = 0.25, P(B_4) = 0.40$$
$$P(A | B_1) = 0.50, P(A | B_2) = 0.60, P(A | B_3) = 0.80, P(A | B_4) = 0.85$$

By the **Law of Total Probability**:

$$P(A) = \sum_{i=1}^4 P(B_i) P(A | B_i)$$

```
# Percentage of images supplied by each sensor (as proportions)
P_B <- c(0.15, 0.20, 0.25, 0.40)

# Percentage of relevant images from each sensor (as proportions)
P_A_given_B <- c(0.50, 0.60, 0.80, 0.85)

# Individual contribution of each sensor to overall relevance
contribution <- P_B * P_A_given_B
names(contribution) <- paste("Sensor", 1:4)
contribution
```

```
## Sensor 1 Sensor 2 Sensor 3 Sensor 4
##      0.075      0.120      0.200      0.340
```

```
# Overall probability (percentage) of relevant images
P_A <- sum(contribution)
P_A
```

```
## [1] 0.735
```

```
cat("Overall percentage of relevant images =", P_A * 100, "%\n")
```

```
## Overall percentage of relevant images = 73.5 %
```

Answer.

$$P(A) = (0.15)(0.50) + (0.20)(0.60) + (0.25)(0.80) + (0.40)(0.85)$$

$$P(A) = 0.075 + 0.12 + 0.20 + 0.34 = 0.735$$

The overall percentage of relevant images is **73.5%**.

2. Problem (#6)

A fair coin is tossed twice. Let

- E_1 = event that both tosses have the same outcome, $E_1 = \{HH, TT\}$
- E_2 = event that the first toss is a head, $E_2 = \{HH, HT\}$
- E_3 = event that the second toss is a head, $E_3 = \{TH, HH\}$

Show that E_1, E_2, E_3 are **pairwise independent** but **not mutually independent**.

Solution The sample space for two coin tosses is

$$S = \{HH, HT, TH, TT\}$$

with each outcome equally likely, i.e. probability 1/4 each.

```
# Sample space
S <- c("HH", "HT", "TH", "TT")
p_outcome <- 1/4 # each outcome equally likely

# Define the events
E1 <- c("HH", "TT") # both tosses same
E2 <- c("HH", "HT") # first toss is head
E3 <- c("TH", "HH") # second toss is head

P_E1 <- length(E1) * p_outcome
P_E2 <- length(E2) * p_outcome
P_E3 <- length(E3) * p_outcome

c(P_E1 = P_E1, P_E2 = P_E2, P_E3 = P_E3)
```

```
## P_E1 P_E2 P_E3
## 0.5 0.5 0.5
```

Each event has probability 1/2:

$$P(E_1) = P(E_2) = P(E_3) = \frac{2}{4} = \frac{1}{2}$$

Pairwise independence

Two events X and Y are independent if $P(X \cap Y) = P(X)P(Y)$.

```
# Intersections
E1_E2 <- intersect(E1, E2)
E1_E3 <- intersect(E1, E3)
E2_E3 <- intersect(E2, E3)

P_E1E2 <- length(E1_E2) * p_outcome
P_E1E3 <- length(E1_E3) * p_outcome
P_E2E3 <- length(E2_E3) * p_outcome

data.frame(
  Pair = c("E1 & E2", "E1 & E3", "E2 & E3"),
  Intersection = c(paste(E1_E2, collapse=","),
                   paste(E1_E3, collapse=","),
                   paste(E2_E3, collapse=",")),
  P_intersection = c(P_E1E2, P_E1E3, P_E2E3),
  Product_of_P = c(P_E1*P_E2, P_E1*P_E3, P_E2*P_E3)
)
```

```
##      Pair Intersection P_intersection Product_of_P
## 1 E1 & E2          HH           0.25         0.25
## 2 E1 & E3          HH           0.25         0.25
## 3 E2 & E3          HH           0.25         0.25
```

- $E_1 \cap E_2 = \{HH\} \Rightarrow P(E_1 \cap E_2) = \frac{1}{4} = P(E_1)P(E_2) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$
- $E_1 \cap E_3 = \{HH\} \Rightarrow P(E_1 \cap E_3) = \frac{1}{4} = P(E_1)P(E_3) = \frac{1}{4}$
- $E_2 \cap E_3 = \{HH\} \Rightarrow P(E_2 \cap E_3) = \frac{1}{4} = P(E_2)P(E_3) = \frac{1}{4}$

Since all three pairwise products match the corresponding joint probabilities, E_1, E_2, E_3 are **pairwise independent**.

Mutual independence check

For **mutual** independence, in addition to the pairwise condition we also need

$$P(E_1 \cap E_2 \cap E_3) = P(E_1)P(E_2)P(E_3)$$

```

E123 <- intersect(intersect(E1, E2), E3)
P_E123 <- length(E123) * p_outcome
product_all <- P_E1 * P_E2 * P_E3

data.frame(
  Quantity = c("P(E1 E2 E3)", "P(E1)P(E2)P(E3)"),
  Value = c(P_E123, product_all)
)

```

```

##           Quantity Value
## 1 P(E1 E2 E3) 0.250
## 2 P(E1)P(E2)P(E3) 0.125

```

$$E_1 \cap E_2 \cap E_3 = \{HH\} \Rightarrow P(E_1 \cap E_2 \cap E_3) = \frac{1}{4}$$

$$P(E_1)P(E_2)P(E_3) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$$

Since

$$\frac{1}{4} \neq \frac{1}{8},$$

the mutual-independence condition **fails**.

Conclusion E_1, E_2, E_3 satisfy $P(E_i \cap E_j) = P(E_i)P(E_j)$ for every pair (i, j) , so they are **pairwise independent**. However, because $P(E_1 \cap E_2 \cap E_3) \neq P(E_1)P(E_2)P(E_3)$ ($\frac{1}{4} \neq \frac{1}{8}$), the three events are **not mutually independent**.